Event Boundaries and Recognition Memory

Mnemonic Similarity Task — BRSM Final Report

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Abstract

We experience continuous sensory input, yet the mind organises experience into discrete events separated by *event boundaries*. According to Event Segmentation Theory, the cognitive system consolidates the departing event at each transition, leaving a brief attentional gap for the incoming event while enhancing encoding fidelity just before the boundary. We tested these predictions using the Mnemonic Similarity Task (MST) with 161 participants assigned to three between-subjects groups: a compound boundary (**Both**: item-class shift + task shift; $n = 52$), item-class shift only (**Item-Only**; $n = 56$), and task shift only (**Task-Only**; $n = 53$). *Phase 1* confirmed a significant Post-Boundary impairment in target recognition exclusively in the Both group ($t(51) = -3.47$, $p = 0.0005$, $d \approx 0.48$), directly replicating Morse et al. (2023). *Phase 2* revealed that this impairment was driven entirely by Object stimuli, which appear at the leading edge of each new event block (boundary position × stimulus class: $F(1, 51) = 5.995$, $p = 0.018$). A Pre-Boundary lure-discrimination enhancement was not replicated at any similarity-bin level, attributable to a reduced Foil pool. Results support a stimulus-category-specific account of post-boundary encoding disruption and confirm that a compound boundary is necessary for item-level memory effects.

1 Introduction

1.1 Event Segmentation and Memory

Our lives unfold as a continuous stream of information, yet we understand and remember experience in terms of discrete events (Zacks & Tversky, 2001; Zacks et al., 2007). Event Segmentation Theory (EST; Zacks et al., 2007) proposes that the cognitive system maintains a predictive event model that is updated at *event boundaries* — moments when perceptual predictions fail. At each boundary, the system consolidates the current event into long-term memory and resets to encode the incoming event (Radvansky, 2017), theoretically creating a brief attentional gap for the boundary-adjacent item and driving the hippocampus into a high-fidelity encoding state for material at the event's close. A reliable behavioural signature is a slowdown in encoding reaction times immediately following a boundary (DuBrow & Davachi, 2013), reflecting the cognitive cost of event-model updating.

Memory for individual events is further shaped by hippocampal *pattern separation* — the ability to store similar experiences as non-overlapping neural traces (Yassa & Stark, 2011). Recent neuroimaging studies demonstrate that hippocampal activity peaks at event boundaries (Baldassano et al., 2017; Ben-Yakov & Henson, 2018; Reagh et al., 2020), linking boundary-elicited consolidation to the formation of distinct, high-fidelity episodic traces.

The Mnemonic Similarity Task (MST; Stark et al., 2019) exploits hippocampal pattern separation using three trial types: **Targets** (exact repeats), **Lures** (perceptually similar, graded across five similarity bins), and **Foils** (novel images). Two bias-corrected indices quantify memory fidelity:

$$\text{REC} = P(\text{“old”} \mid \text{Target}) - P(\text{“old”} \mid \text{Foil}) \tag{1}$$

$$\text{LDI} = P(\text{“similar”} \mid \text{Lure}) - P(\text{“similar”} \mid \text{Foil}) \tag{2}$$

Values above zero indicate above-chance performance relative to the Foil baseline.

1.2 Boundary Effects on Recognition Memory and The Present Study

Morse et al. (2023) first combined event-boundary manipulations with the MST, finding a directional asymmetry: Post-Boundary items showed impaired target recognition (REC<0 vs. Non-Boundary), while Pre-Boundary items showed enhanced lure discrimination (LDI>0 vs. Non-Boundary). Critically, three experiments showed that only a *compound* boundary — combining item-class shift and task shift — produced these effects; either component alone was insufficient, though task-shift boundaries alone did elicit an RT slowdown.

The present study replicates Morse et al. (2023) using three between-subjects groups: **Both** (compound boundary), **Item-Only** (class shift only), and **Task-Only** (task shift only). Key procedural differences: the object encoding question was changed to “Will it fit in a shoebox?”; events consisted of 7 images across 40 blocks (280 encoding trials); the recognition test used an *old/new/similar* format with 60 Targets, 60 Lures, and 30 Foils.

Hypotheses. (H1a) Post-Boundary REC will be impaired relative to Mid-Boundary in the Both group; (H1b) Pre-Boundary LDI will be enhanced relative to Mid-Boundary in the Both group. Item-Only and Task-Only are expected to show weaker or absent effects.

2 Methods

2.1 Participants

A total of 161 participants completed both an encoding phase and a recognition test: Item-Only ($n = 56$), Task-Only ($n = 53$), and Both ($n = 52$). No exclusion criteria were applied beyond data completeness.

2.2 Design

A 3 × 3 mixed design was used. The between-subjects factor was boundary type; the within-subjects factor was boundary position. Items at serial position 0 within each 7-item event ($\text{trial}_n \bmod 7 = 0$) were labelled **Post-Boundary**; items at position 6 were **Pre-Boundary**; positions 1–5 formed the **Mid-Boundary** baseline. Each block thus contributed one Post-Boundary trial, one Pre-Boundary trial, and five Mid-Boundary trials.

2.3 Stimuli and Procedure

During encoding, each image was displayed for 3 seconds. Participants responded to a categorical judgement: *“Will it fit in a shoebox?”* for Objects and *“Indoor or outdoor?”* for Scenes. The recognition test comprised 150 trials: 60 Targets (20 per position), 60 Lures (20 per position, across 5 similarity bins), and 30 Foils. Participants responded *old*, *new*, or *similar*.

2.4 Data Processing

Encoding RT was computed as: RT = `key_resp_9.rt` if the participant responded before image offset; RT = 3000 ms + `key_resp_8.rt` if after offset. Four files missing `key_resp_8.rt` had this column added as NA. Lure bin assignments were read from external mapping files

(`Set6 bins.txt` for Objects; `SetScC bins.txt` for Scenes). All analyses were conducted in R (`tidyverse`, `rstatix`).

2.5 Statistical Analysis

Phase 1. Difference scores (boundary position − Mid-Boundary) were computed per participant. One-sample t-tests with directional alternatives tested deviation from zero (Post REC < 0; Pre LDI > 0). Bonferroni correction was applied across four primary tests ($\alpha = 0.05/4 = 0.0125$).

Phase 2. (i) One-way ANOVA compared Post-Boundary REC across groups, preceded by Shapiro-Wilk and Levene's checks; the appropriate parametric or non-parametric test was used depending on outcomes. (ii) Chi-squared goodness-of-fit tested whether the proportion of Both- group participants showing the expected direction exceeded 50% chance. (iii) A 2 × 2 repeated-measures ANOVA (boundary position × stimulus class) was conducted in the Both group. (iv) A mixed ANOVA (boundary position × lure bin) examined whether any Pre-Boundary LDI advantage was concentrated in high-similarity bins.

3 Results — Phase 1 (Recap)

3.1 Raw Response Proportions

Foil false-alarm rates were consistently low (Item-Only: 3.5%; Task-Only: 3.1%; Both: 4.7%), confirming valid bias correction. In the Both group, the Post-Boundary target hit rate (0.596) was 7.2 percentage points below Mid-Boundary (0.668), while Pre-Boundary (0.651) sat in between. Item-Only and Task-Only showed flat profiles across boundary positions. Figure 1 displays mean response proportions by stimulus type and group.

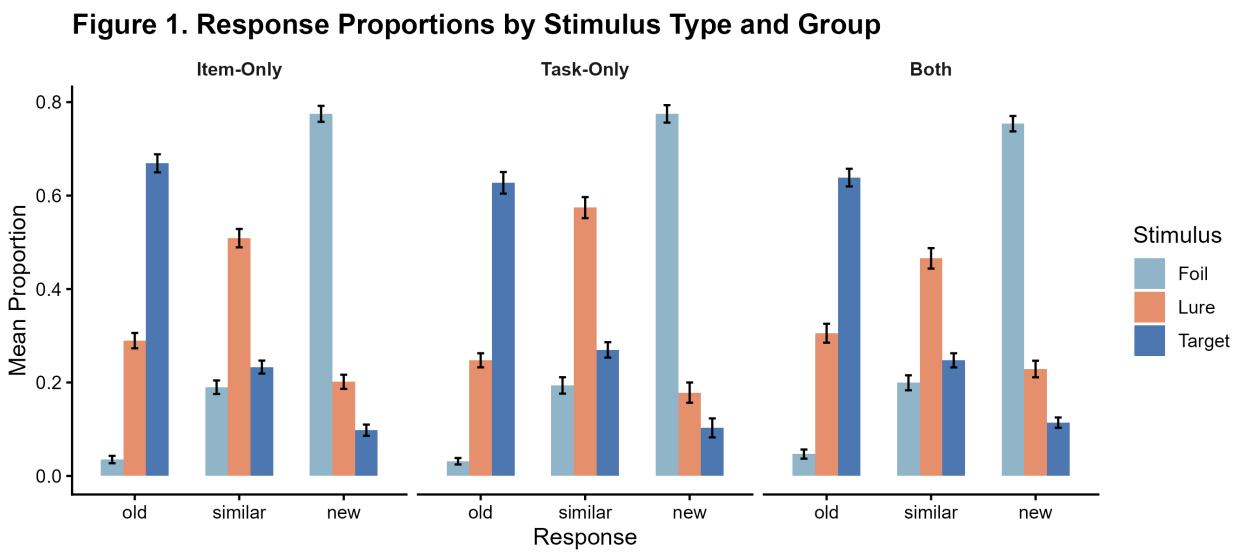

Figure 1: Mean proportion of each response type (*old*, *similar*, *new*) by stimulus type (Foil, Lure, Target) and group. Error bars = ±1 SE. Foil false-alarm rates are consistently low across all groups, confirming adequate task engagement.

3.2 Descriptive Statistics: REC and LDI

Table 1 reports bias-corrected scores. The Both group shows the most pronounced boundary-related pattern: Post-Boundary REC ($M = 0.549$, $SD = 0.164$) was 0.072 units below Mid-Boundary ($M = 0.621$), while Pre-Boundary ($M = 0.604$) sat in between. Item-Only REC was essentially flat (Post: 0.635; Mid: 0.640; Pre: 0.627). For LDI, Both-group values

were consistently lower overall (Post: 0.251; Mid: 0.283; Pre: 0.265) than Item-Only (0.314–0.330) or Task-Only (0.369–0.387). Figure 2 plots bias-corrected REC and LDI by boundary position and group.

Table 1: Bias-corrected REC and LDI by group and boundary position (from CSV data).

Group	Boundary Position	N	REC M	REC SD	LDI M	LDI SD
Item-Only	Post-Boundary	56	0.635	0.188	0.314	0.172
	Mid-Boundary	56	0.640	0.178	0.330	0.188
	Pre-Boundary	56	0.627	0.157	0.314	0.213
Task-Only	Post-Boundary	53	0.563	0.204	0.386	0.229
	Mid-Boundary	53	0.601	0.183	0.369	0.180
	Pre-Boundary	53	0.624	0.195	0.387	0.205
Both	**Post-Boundary**	**52**	**0.549**	**0.164**	**0.251**	**0.185**
	Mid-Boundary	52	0.621	0.175	0.283	0.221
	Pre-Boundary	52	0.604	0.155	0.265	0.184

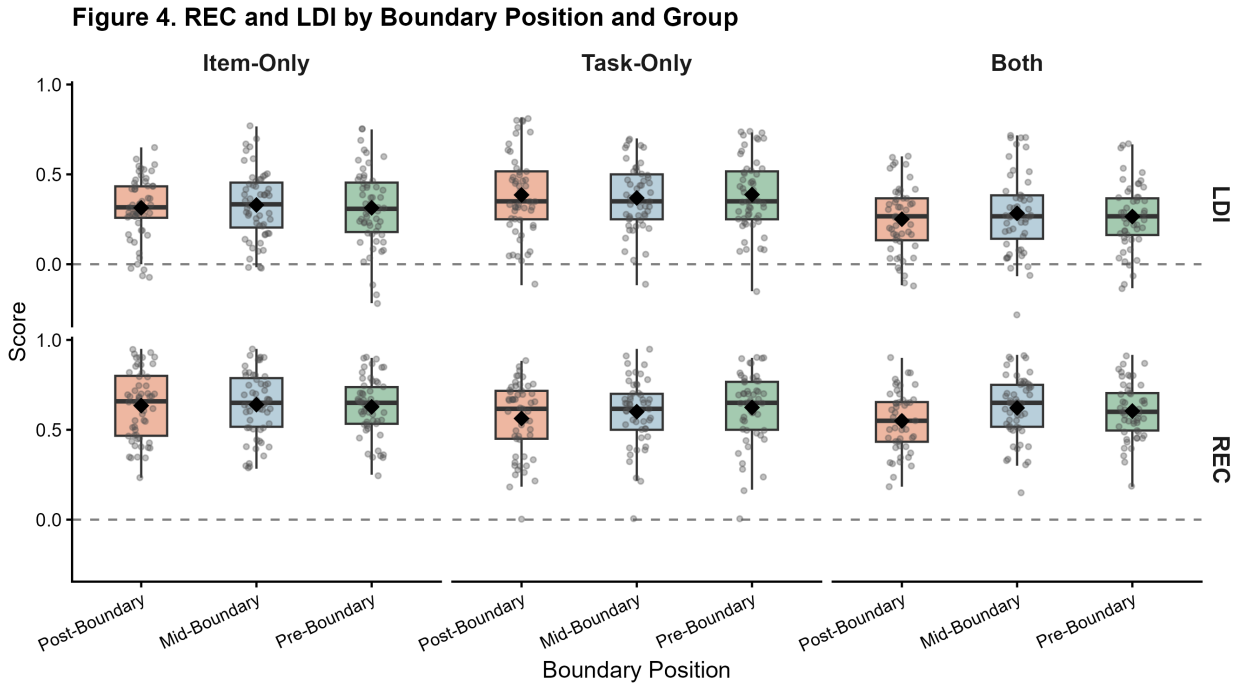

Figure 2: Bias-corrected REC (bottom row) and LDI (top row) by boundary position and group. Dashed line = zero (chance baseline). Diamonds = group means. The Post-Boundary dip in REC is most pronounced in the Both group.

3.3 Encoding RT: Manipulation Check

A Post-Boundary RT slowdown confirms genuine boundary perception (Radvansky, 2017). The Both group showed ≈+217 ms Post-Boundary slowing relative to Mid-Boundary (≈1908 vs. ≈1691 ms); Task-Only showed a similar pattern (+235 ms). Item-Only showed virtually no change (−16 ms), confirming that a stimulus-class shift alone is not perceived as a functional boundary. Figure 3 illustrates these RT patterns.

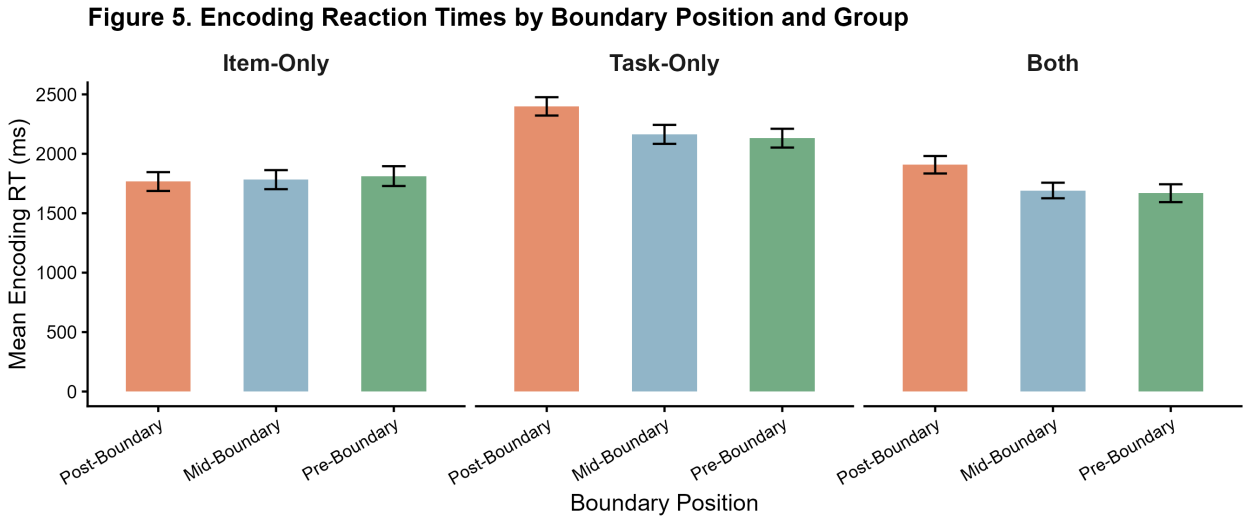

Figure 3: Mean encoding RT (ms) by boundary position and group. Error bars = ± 1 SE. Post-Boundary slowing in Both (+217 ms) and Task-Only (+235 ms) confirms genuine boundary perception; Item-Only shows no such effect (−16 ms).

3.4 One-Sample t-Tests: Did Boundary Position Affect Memory?

Per-participant difference scores (boundary position − Mid-Boundary) were tested against zero (Table 2). Figure 4 plots all difference scores with annotated p-values.

Table 2: One-sample t-tests: difference scores vs. zero. Bonferroni $\alpha = 0.0125$. * = survives correction.

Group	Meas.	Position	N	M	SD	t	p
Both	REC	Post	52	−0.072	0.150	−3.468	**0.0005***
Item-Only	REC	Post	56	−0.005	0.161	−0.249	0.402
Task-Only	REC	Post	53	−0.038	0.153	−1.801	0.039
Both	LDI	Pre	52	−0.018	0.153	−0.859	0.803
Item-Only	LDI	Pre	56	−0.016	0.144	−0.833	0.796
Task-Only	LDI	Pre	53	+0.019	0.138	+0.992	0.163

Both — H1a. $M_{\text{diff}} = -0.072$, $t(51) = -3.468$, $p = 0.0005$, $d \approx 0.48$ (moderate). Survives Bonferroni correction. Post-Boundary REC is reliably impaired in the compound-boundary group, directly replicating Morse et al. (2023).

Both — H1b. $M_{\text{diff}} = -0.018$, $t(51) = -0.859$, $p = 0.803$. Wrong direction; not replicated. The most plausible explanation is the reduced Foil pool (30 vs. 120 in the original), which greatly inflates noise in per-participant LDI estimates.

Item-Only and Task-Only did not reach significance for either measure after Bonferroni correction, consistent with the compound-boundary requirement.

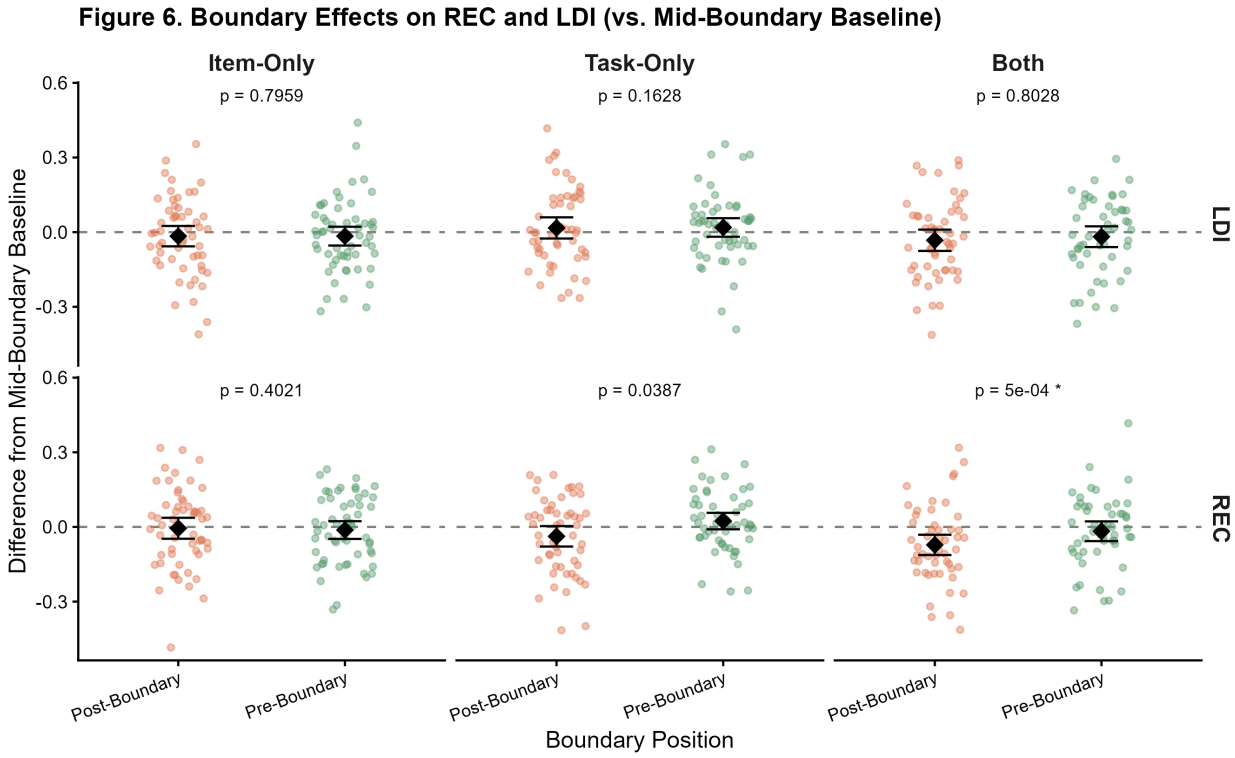

Figure 4: REC and LDI difference scores (vs. Mid-Boundary baseline) by group. Diamonds = group means ±95 % CI; annotated p-values from one-sample t-tests (Bonferroni-corrected). The significant Post-Boundary REC impairment ($p = 5 \times 10^{-4}$, *) is visible only in the Both group.

4 Results — Phase 2 (Extended Analyses)

4.1 Does Boundary Strength Drive the Effect? (One-Way ANOVA)

Normality was satisfied in all groups (Shapiro-Wilk $p > 0.05$) and variances were homogeneous (Levene's $F(2, 158) = 0.326$, $p = 0.722$). The omnibus ANOVA on Post-Boundary REC difference scores was *not* significant. Although the Both group had the most negative mean (-0.072) vs. Item-Only (-0.005) and Task-Only (-0.038), within-group variability precluded a significant between-group difference. The violin plot in Figure 5 illustrates the overlapping distributions.

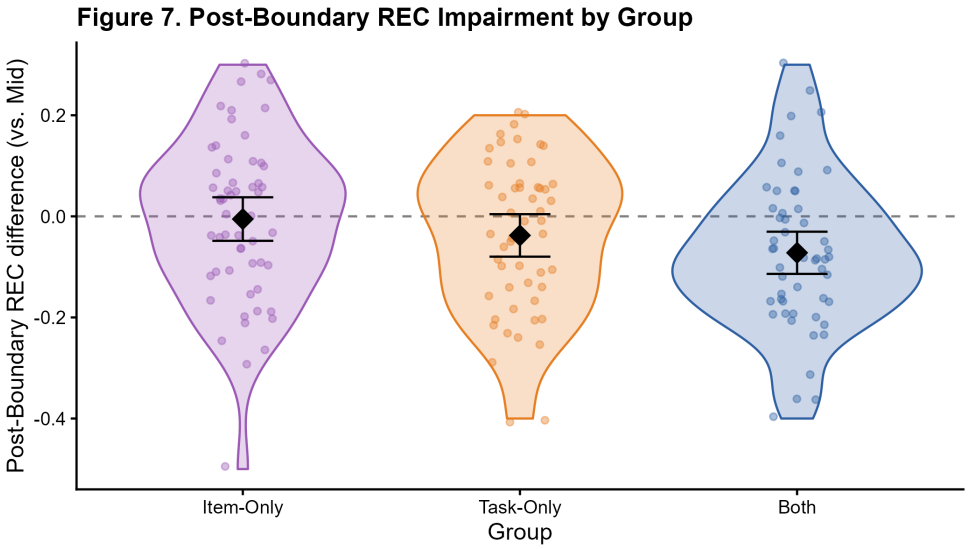

Figure 5: Post-Boundary REC difference scores (vs. Mid-Boundary) by group. Diamonds = means ±95 % CI. Although the Both group's mean is most negative, the one-way ANOVA was not significant given large within-group variability.

4.2 Is the Effect Consistent Across Participants? (Chi-Squared Test)

Table 3 and Figure 6 summarise directional consistency. In the Both group, 59.6% of participants showed the expected direction ($\chi^2(1) = 1.923$, $p = 0.166$). Strikingly, Item-Only showed only 33.9% in the expected direction, significantly *below* chance ($\chi^2(1) = 5.786$, $p = 0.016$), consistent with random memory variability in the absence of a perceived boundary.

Table 3: Chi-squared: proportion with expected REC direction (Post<Pre) vs. 50% chance.

Group	N	% Expected	$\chi^2(1)$	p
Item-Only	56	33.9	5.786	0.016*
Task-Only	53	60.4	2.283	0.131
Both	52	59.6	1.923	0.166

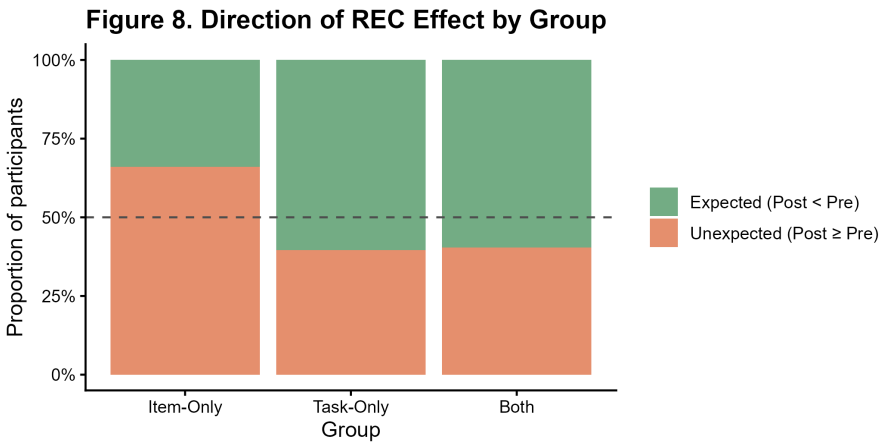

Figure 6: Proportion of participants showing the expected REC direction (Post<Pre, green) vs. unexpected (Post≥Pre, orange) by group. Dashed line = 50% chance. Item-Only is significantly below chance ($p = 0.016$).

4.3 Do Objects and Scenes Differ? (Repeated-Measures ANOVA)

If the Post-Boundary impairment is a general attentional gap, it should affect Objects and Scenes equally. If it is specific to the category at the *leading edge* of the new event, we expect a significant boundary position × stimulus class interaction.

Normality was violated in Object cells ($p < 0.025$) but not Scenes ($p > 0.44$); the RM-ANOVA is robust to moderate violations with $n = 52$.

Table 4: Repeated-measures ANOVA: boundary position × stimulus class on REC (Both group).

Effect	DF_n	DF_d	F	p
Boundary position	1	51	7.959	0.007
Stimulus class	1	51	7.416	0.009
Boundary pos × Stimulus class	**1**	**51**	**5.995**	**0.018**

The interaction (Table 4, Figure 7) was significant. Simple effects showed a boundary effect for Objects ($t(51) = -3.630$, $p < 0.001$) but not Scenes ($t(51) = -0.425$, $p = 0.673$). The Post-Boundary impairment is therefore **driven entirely by Object stimuli**: in the Both group's alternating design, Objects always occupy the first slot of each new event block and bear the full attentional cost of the event-model update, while Scenes appear mid-event and encode normally.

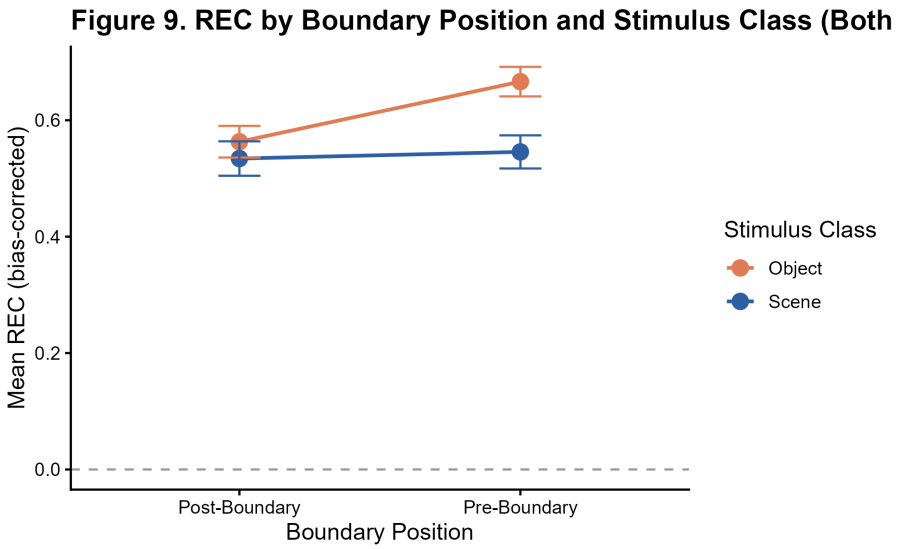

Figure 7: Mean bias-corrected REC for Object and Scene stimuli at Post- and Pre-Boundary positions (Both group, $n = 52$). Error bars = ± 1 SE. The boundary effect is significant for Objects ($p < 0.001$) but absent for Scenes ($p = 0.673$).

4.4 Is the LDI Null a Power Problem? (Lure Bin Analysis)

With only 30 Foils, each bin has ≈4 lure trials per boundary position per participant. If a Pre-Boundary advantage exists, it should be strongest at Bin 1 (highest pattern-separation demand). A mixed ANOVA (boundary position × lure bin) in the Both group (Table 5) showed a significant main effect of bin ($F(4, 376) = 6.413$, $p < 0.001$), confirming the expected monotonic increase in LDI from Bin 1 to Bin 5. However, neither the boundary position main

effect ($F(1, 94) = 0.083$, $p = 0.774$) nor the interaction ($F(4, 376) = 1.317$, $p = 0.263$) was significant (Figure 8). The LDI null is therefore *not* a bin-specific power failure.

Table 5: Mixed ANOVA: boundary position × lure bin on LDI (Both group).

Effect	DF_n	DF_d	F	p
Boundary position	1	94	0.083	0.774
Lure bin	**4**	**376**	**6.413**	**<0.001**
Boundary pos × Lure bin	4	376	1.317	0.263

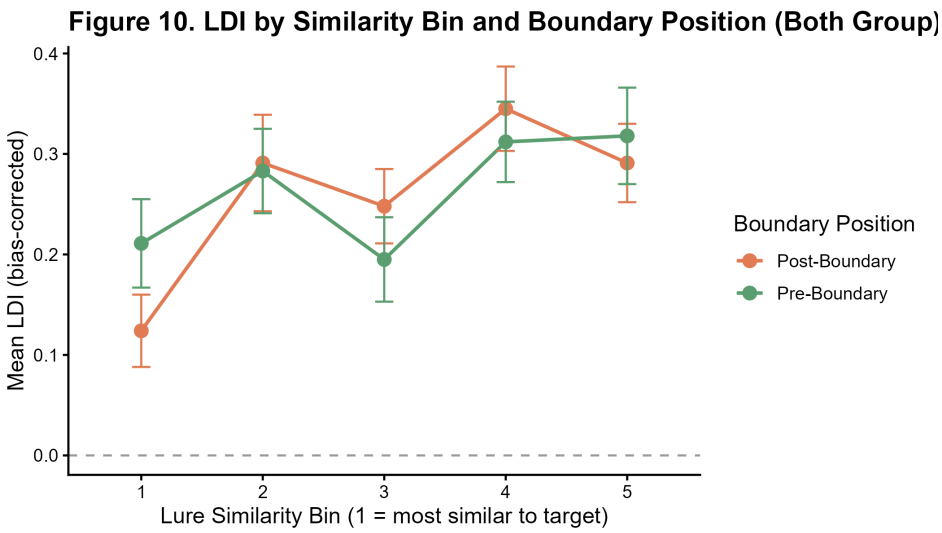

Figure 8: Mean LDI by lure similarity bin and boundary position (Both group). Bin 1 = most similar (hardest pattern separation); Bin 5 = least similar. Error bars = ±1 SE. No significant boundary position × bin interaction was found ($p = 0.263$).

5 Discussion

5.1 Summary of Findings

Phase 1 established the central result: items encoded immediately after a compound event boundary are recognised significantly less accurately than Mid-Boundary items ($t(51) = -3.47$, $p = 0.0005$, $d \approx 0.48$), replicating Morse et al. (2023). Encoding RT confirmed genuine boundary perception in Both (+217 ms) and Task-Only (+235 ms) but not Item-Only (−16 ms). The Pre-Boundary LDI enhancement was not replicated, most plausibly due to the reduced Foil pool (30 vs. 120).

Phase 2 yielded four key results: (i) The one-way ANOVA was non-significant, indicating the Both group impairment was not reliably larger than other groups given within-group variability. (ii) Only 59.6% of Both-group participants showed the expected direction, reflecting substantial individual variability in susceptibility to boundary-driven encoding disruption. (iii) The Post-Boundary impairment was driven entirely by Object stimuli ($F(1, 51) = 5.995$, $p = 0.018$), confirming a category-specific rather than a general positional effect. (iv) Bin-level LDI analyses ruled out a power-based account of the null Pre-Boundary LDI finding.

5.2 Theoretical Implications

These results support and extend the hippocampal snapshot-encoding account of Morse et al. (2023). The Post-Boundary impairment in the compound-boundary group, its absence

in single-component groups, and the category specificity are all consistent with the view that event boundaries drive hippocampal consolidation of the departing event at the cost of encoding the incoming one (Radvansky, 2017; Yassa & Stark, 2011; Sols et al., 2017). The category-specific finding is novel relative to Morse et al. and suggests that the boundary-onset "attentional blink" (cf. Dux & Marois, 2009) is organised around the identity and structural role of the first item in the new event, not temporal proximity alone. This distinction may inform future designs seeking to disentangle temporal and categorical contributions to boundary-related memory effects.

5.3 Limitations and Future Directions

Reduced Foil pool: The reduction from 120 to 30 Foils is the most consequential design departure. It degrades LDI precision globally and makes bin-level analyses underpowered. Future replications should restore the original count.

Between-subjects boundary manipulation: Individual-difference confounds could be eliminated by a within-subjects design exposing the same participant to multiple boundary types.

Object-first confound: Objects always appeared first in each new event block in the Both group. Future studies should counterbalance which category appears at boundary onsets to determine whether the impairment follows category identity or position.

Online testing: Like the original study, this experiment was run online, introducing variability in attention and screen conditions. In-laboratory replications would strengthen confidence in the effects.

Individual Contributions

Member	Contributions
T. Sathvik Reddy	Data loading & preprocessing pipeline, REC/LDI computation, one-sample t-tests, one-way ANOVA & chi-squared analyses.
Y. Havinash Reddy	Visualisations & figure preparation, encoding RT analysis, lure bin mixed ANOVA (Phase 2).
D. Subhash Chandra	Statistical testing, normality checks (Shapiro–Wilk, Levene's test), RM-ANOVA (Phase 2), report writing & formatting.

All members contributed collaboratively to experimental design, interpretation, and writing.

Code: https://github.com/SubhashChandraDangeti/BRSM_MST

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